Red Team Week 1

Infrastructure Build Report

Objective

I deployed a segmented, firewalled network architecture (Management, Red Team, Target) using VirtualBox, pfSense, Kali Linux, and Metasploitable 2 to meet the Week 1 requirements.

1. Prerequisites and Environment

I used my dual-booted Kali Linux machine as the host OS and installed VirtualBox as the hypervisor. I downloaded the required ISOs for pfSense CE v2.7.x and Metasploitable 2. I ran the commands below in my Kali terminal to download the necessary files.

Downloading pfSense CE 2.7.2

```
cd ~/Downloads
wget -q --show-progress -O - https://atxfiles.netgate.com/mirror/downloads/pfSense-CE-2.7.2-RELEASE-amd64.iso.gz | gzip -d -c > pfSense-CE-2.7.2-RELEASE-amd64.iso
```

Downloading and extracting Metasploitable 2

```
wget
https://downloads.sourceforge.net/project/metasploitable/Metasploitable2/Metasploitable2-Linux.zip
unzip Metasploitable2-Linux.zip
```

2. VirtualBox Network Infrastructure (Phase 1)

I configured the virtual network wires before starting any VMs.

A. Host-Only Networks

I discovered that VirtualBox restricts Host-Only IPs by default, so I allowed custom ranges by running:

```
echo "* 10.0.0.0/8 192.168.0.0/16" | sudo tee /etc/vbox/networks.conf
```

I created the Management Network (vboxnet0):

```
sudo VBoxManage hostonlyif create
sudo VBoxManage hostonlyif ipconfig vboxnet0 --ip 192.168.1.1 --netmask 255.255.255.0
```

I created the Red Team Network (vboxnet1):

```
sudo VBoxManage hostonlyif create
sudo VBoxManage hostonlyif ipconfig vboxnet1 --ip 10.10.10.1 --netmask 255.255.255.0
```

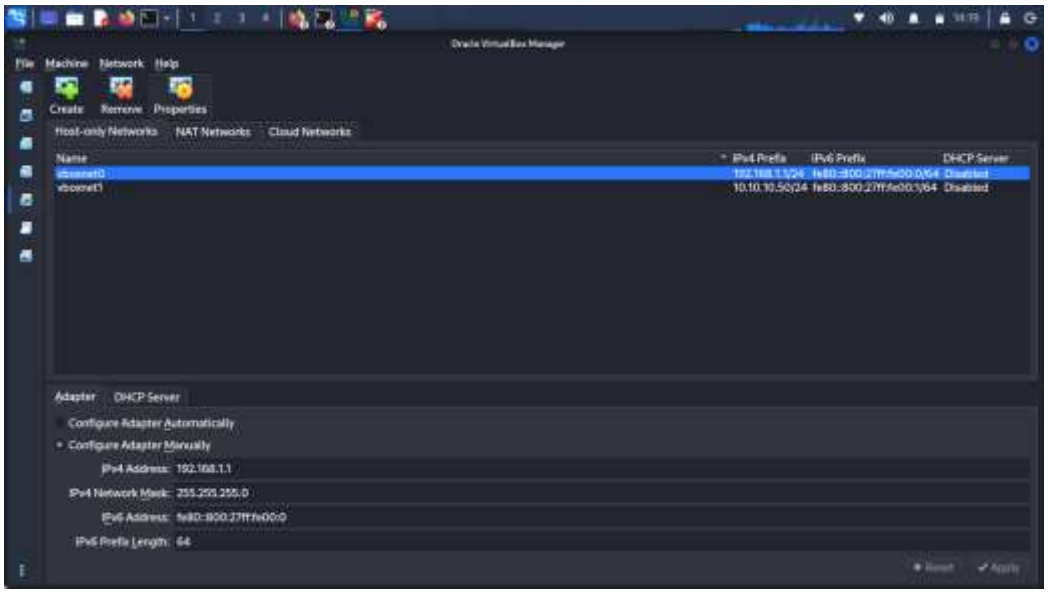


Figure 1: VirtualBox Host-only Networks showing vboxnet0 and vboxnet1

B. VM Adapter Assignments

I opened the VirtualBox GUI and assigned the adapters as follows.

VM	Adapter	Type	Attached To
pfSense	Adapter 1 (WAN)	Bridged Adapter	Host Wi-Fi
pfSense	Adapter 2 (LAN)	Host-only Adapter	vboxnet0
pfSense	Adapter 3 (OPT1)	Host-only Adapter	vboxnet1
pfSense	Adapter 4 (OPT2)	Internal Network	Target_Net
Metasploitable	Adapter 1	Internal Network	Target_Net

3. Operating System Configurations (Phase 2)

A. pfSense Router Setup

I booted pfSense and assigned the interfaces at the console by selecting option 1:

- WAN = em0
- LAN = em1
- OPT1 = em2
- OPT2 = em3

I then set the IPs using Console Option 2:

- LAN (Management): 192.168.1.2/24, DHCP disabled
- OPT1 (Red Team): 10.10.10.2/24, DHCP disabled

- OPT2 (Target): 10.20.20.1/24, DHCP disabled

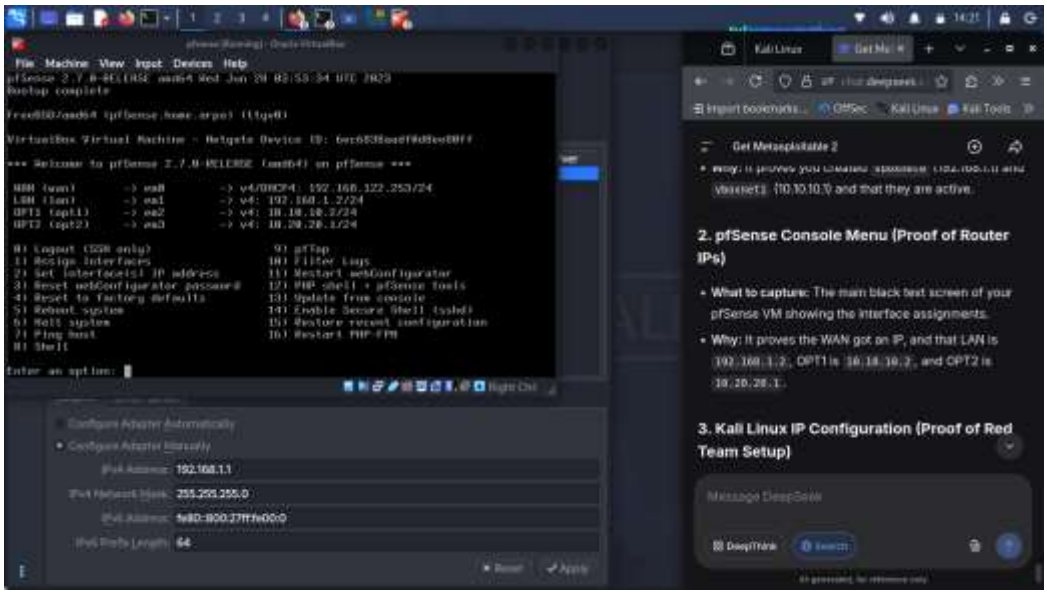


Figure 2: pfSense console showing the WAN, LAN, OPT1 and OPT2 interface assignments

B. Metasploitable 2 Target Setup

I logged into Metasploitable using msfadmin/msfadmin and edited the network file by running:

```
sudo nano /etc/network/interfaces
```

I configured it exactly as:

```
auto eth0
iface eth0 inet static
address 10.20.20.100
netmask 255.255.255.0
gateway 10.20.20.1
```

I applied the settings by running:

```
sudo /etc/init.d/networking restart
```

4. Host Kali Routing (Phase 3)

Since Kali is my physical host, I manually attached it to the virtual networks by running these commands in the Kali terminal.

Management IP

```
sudo ip addr add 192.168.1.100/24 dev vboxnet0
sudo ip link set vboxnet0 up
```

Red Team IP

```
sudo ip addr del 10.10.10.1/24 dev vboxnet1 2>/dev/null
sudo ip addr add 10.10.10.50/24 dev vboxnet1
sudo ip link set vboxnet1 up
```

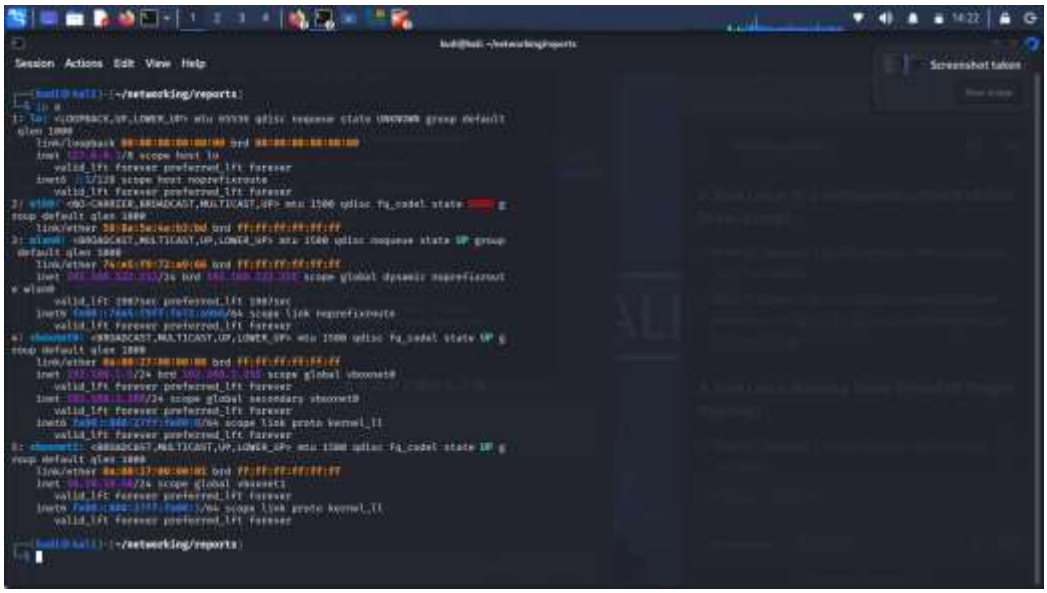


Figure 3: Output of ip a on Kali showing 192.168.1.100 on vboxnet0 and 10.10.10.50 on vboxnet1

Route to the Target network via pfSense

```
sudo ip route add 10.20.20.0/24 via 10.10.10.2 dev vboxnet1
```

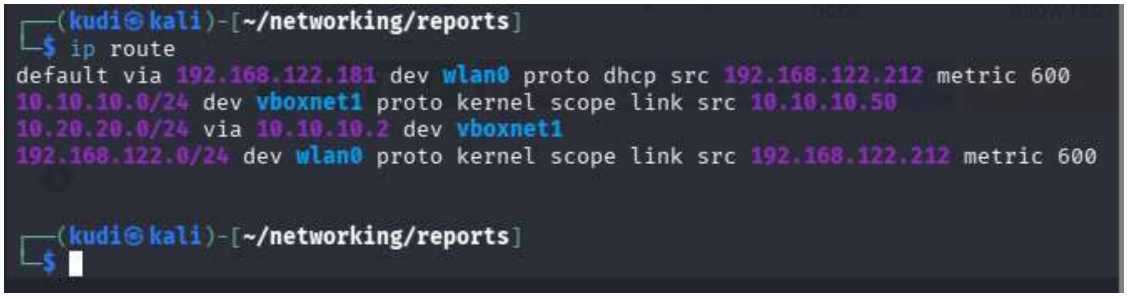


Figure 4: Kali routing table (ip route) showing 10.20.20.0/24 routed via 10.10.10.2

5. Firewall Authorization (Phase 4)

Traffic was initially blocked by pfSense's default deny policy. I logged into the pfSense WebGUI from my Kali browser at https://192.168.1.2 using admin/pfsense.

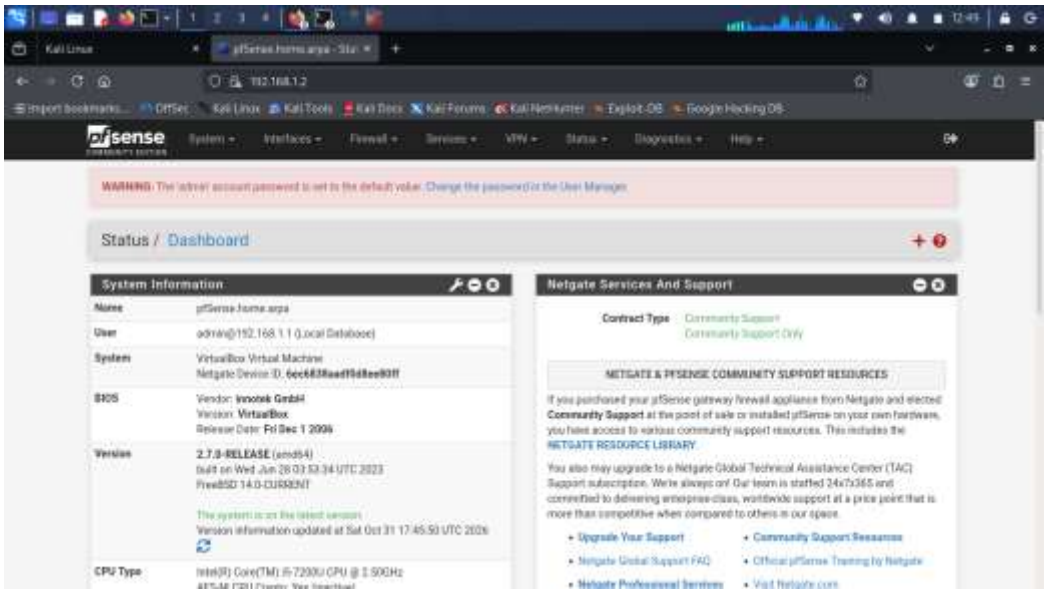


Figure 5: pfSense WebGUI dashboard accessed from Kali at 192.168.1.2

I navigated to Firewall → Rules → OPT1 (Red Team), clicked Add, and configured the rule:

- Action: Pass
- Protocol: Any
- Source: any
- Destination: any

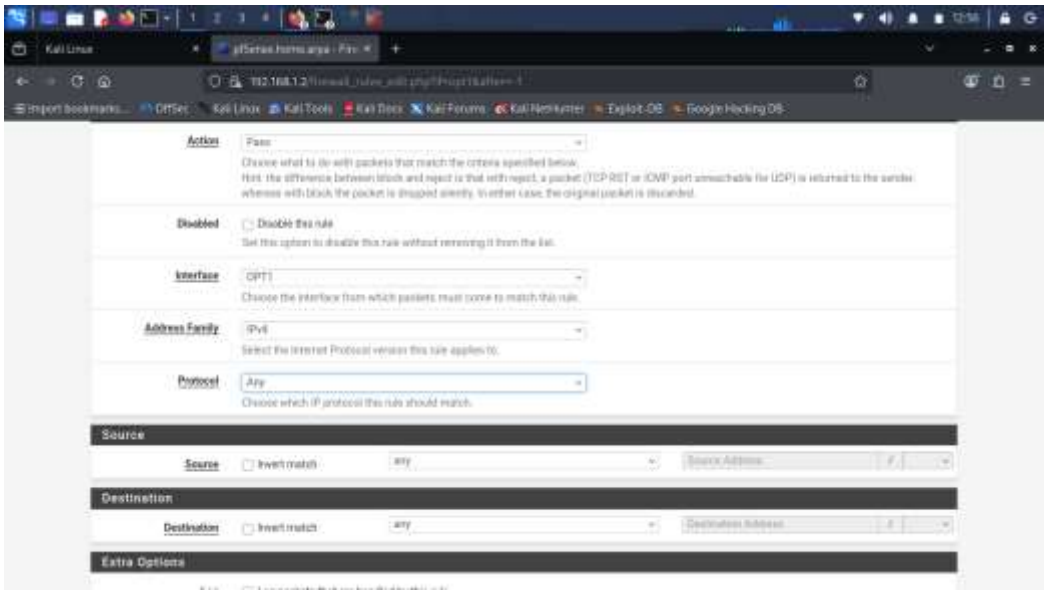


Figure 6: Firewall rule configuration on the OPT1 (Red Team) interface

I clicked Save, and then clicked Apply Changes, which was the critical step to make the rule active.

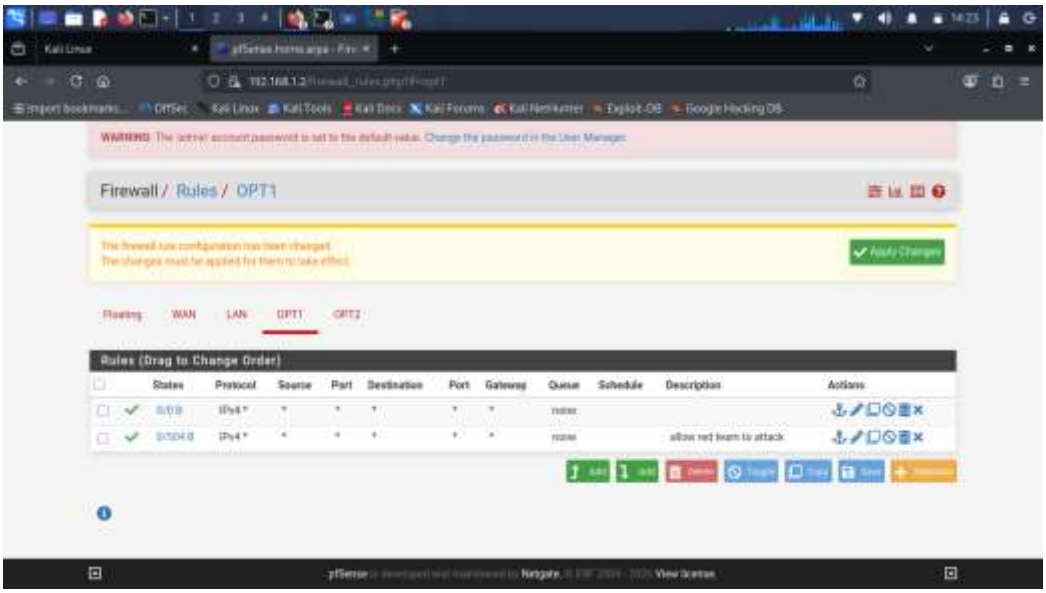


Figure 7: OPT1 rules list with the Pass rule saved and the Apply Changes prompt

6. Validation

From my Kali terminal, I tested the connectivity.

Gateway connectivity

```
ping -c 3 10.10.10.2
```

Target connectivity

```
ping -c 3 10.20.20.100
```

Target port exposure

```
nmap -sV 10.20.20.100
```

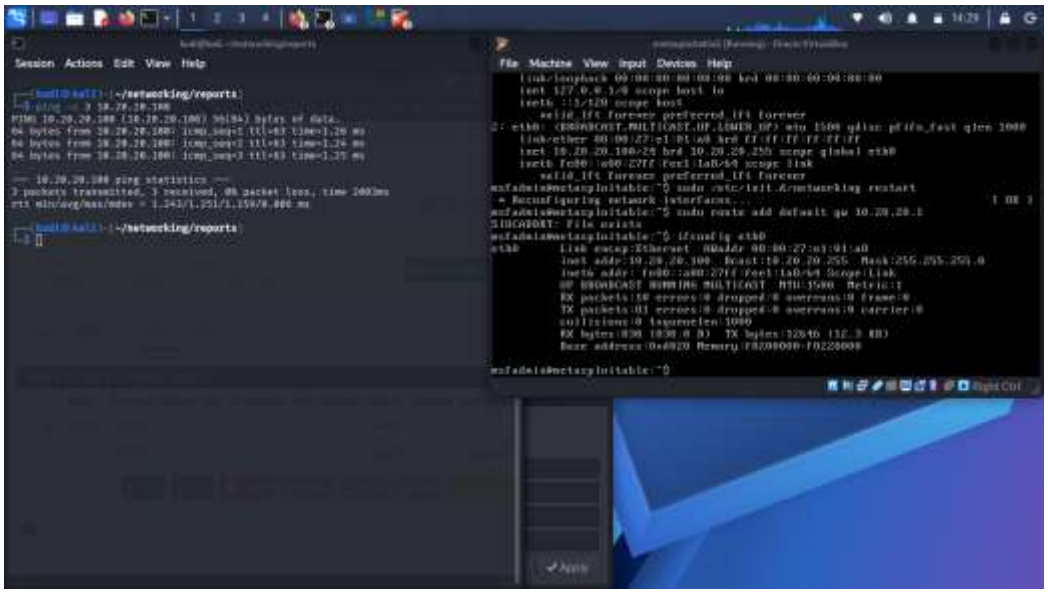


Figure 8: Successful ping from Kali to 10.20.20.100, with the Metasploitable 2 eth0 configuration (right)

Result

The pings returned replies and Nmap showed open ports (FTP, SSH, HTTP, etc.). The Week 1 network infrastructure is 100% operational.

Maintenance Note

> Because Kali is my host OS, the vboxnet IP assignments are not persistent across reboots. If my Kali machine restarts, I must re-run the commands in Phase 3 to restore connectivity.